

Question 1

Let $D(n)$ be the position at which state n differs from goal g , and let $r(w)$ be the nonnegative rarity of dictionary word w . The submitted program uses the following heuristics:

- Steps: $h(n) = |D(n)|$. Each action changes just one position. Every differing position must change at least once before reaching g . Since each action costs 1, every solution costs at least $|D(n)|$. At g the value is 0.
- Scrabble: $h(n) = \text{sum over } i \text{ in } D(n) \text{ of } s(g[i])$, where $s(c)$ is the specified replacement-letter score. Each incorrect position must eventually receive its goal letter, incurring $s(g[i])$ at least once. Other changes have a positive cost. Therefore this sum cannot exceed the true remaining cost. The empty sum at g is 0.
- Frequency: $h(n) = |D(n)|$. Each differing position requires at least one action, and each action costs $1 + r(\text{result}) \geq 1$. Hence the remaining cost is at least $|D(n)|$, with $h(g) = 0$. This heuristic deliberately ignores nonnegative rarity. It is nontrivial for every non-goal state.

Question 2

Word pair	Measure	UCS cost	UCS expanded	A* cost	A* expanded	Reduction
WHY to ASK	Steps	5	773	5	16	97.93%
WHY to ASK	Scrabble	9	765	9	13	98.30%
WHY to ASK	Frequency	38.85540652	808	38.85540652	755	6.56%
BEAR to DUCK	Steps	4	691	4	6	99.13%
BEAR to DUCK	Scrabble	11	2961	11	6	99.80%
BEAR to DUCK	Frequency	38.2402284	1084	38.2402284	720	33.58%

Scrabble gives the largest reduction for both word pairs at 98.30% and 99.80%. Frequency is least informative for both pairs, with 6.56% and 33.58%. The steps heuristic accounts for each required letter correction, while the Scrabble heuristic additionally prices the required goal letters. The frequency heuristic assigns only 1 per mismatch and ignores potentially substantial rarity costs. Many candidate paths consequently have similar low estimates. The counts, rather than runtime, support these comparisons.

Question 3

- Steps - yes: Reversing a legal path reverses every one-letter action and preserves its number of actions. Taking the minimum in each direction therefore gives equal optimal costs.
- Scrabble - no: DOG and COG occur in the supplied dictionary and are adjacent. DOG to COG inserts c (cost 3). COG to DOG inserts d (cost 2). These direct paths are optimal. Every route must eventually set the first character to its goal letter and pay at least its score. Thus the optimal costs differ. More generally, on any path the forward cost minus its reverse cost equals the sum of letter scores of the goal minus that of the start.
- Frequency - no: DOG and COG are adjacent. The supplied rarity values are $r(\text{dog})=5.819885$ and $r(\text{cog})=11.393166$. Direct costs are $1+r(\text{cog})=12.393166$ from DOG to COG and $1+r(\text{dog})=6.819885$ in reverse. More generally, for any path $w_0, \dots, w_k$ and its reverse, the difference is $r(w_k) - r(w_0)$, because all interior rarity terms cancel. Reversing paths gives a bijection, so optimal costs in opposite directions differ by precisely this endpoint difference. A reversible action need not have symmetric cost.

Question 4

The following prompt was given to OpenAI's ChatGPT-6 Sol Light:

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary. The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

- AI Proposal: Add the rarity of the goal to the Hamming-distance baseline for every non-goal state. In symbols, $h_1(g)=0$ and $h_1(n)=|D(n)|+r(g)$ for $n \neq g$. The implementation in `dc.py` retains h_0 for Question 2; h_1 is evaluated separately here, without changing the submitted heuristic.
- Verification: For any non-goal n , at least $|D(n)|$ actions are necessary, contributing at least $|D(n)|$ to frequency cost. The final action necessarily enters g and contributes $r(g)$, which is counted once within the actions already counted. All other rarity terms are nonnegative. Thus every goal-reaching path has cost at least $|D(n)|+r(g)$. At the goal $h_1=0$, proving admissibility. Since $r(g) \geq 0$, $h_1 \geq h_0$ everywhere. For both assigned goals $r(g) > 0$ ($r(\text{ask})=4.2326436$).

$r(\text{duck})=7.2297187$), so the inequality is strict at all non-goal states. No repair is needed.

Frequency Problem	Heuristic	A* cost	Nodes expanded
WHY to ASK	$h_0: D $	38.85540652	755
WHY to ASK	$h_1: D + r(\text{goal})$	38.85540652	634
BEAR to DUCK	$h_0: D $	38.2402284	720
BEAR to DUCK	$h_1: D + r(\text{goal})$	38.2402284	302

The verified heuristic retains optimal costs and expands fewer nodes on both problems, with 121 fewer and 418 fewer respectively. These two experiments support using h_1 here.

AI Disclosure

I used OpenAI's ChatGPT-6 Sol Light with assistance in implementing [dc.py](#) due to my limited experience with Python. Running [dctest.py](#) was a major use of verifying ChatGPT's assistance.